

From Tangential Phase Transport to Normal Asymmetry in Semiprime Shells

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1. Introduction

Recent work, [Inter Scale Confirmation of Angular Pendulum](#), has introduced a dyadic angular representation of semiprime integers, in which an integer

$$n = p \cdot q$$

is embedded on a circular shell indexed by its dyadic scale K_n , and its prime factors are analyzed through associated angular observables. Within this framework, a first set of results established the existence of a structured *tangential phase transport*: the angular position of a semiprime n constrains, in a statistically stable manner, the angular positions of its prime factors across multiple dyadic scales. These results, documented in the preceding study, were obtained without recourse to logarithmic phases and were shown to persist from moderate to large scales.

However, tangential observables alone do not fully characterize the geometric organization induced by the semiprime structure. While tangents capture local angular transport along the shell, they remain insensitive to a deeper directional organization that emerges when the geometry is examined relative to the normal directions of the dyadic circles. This limitation motivates the introduction of a new class of observables: **angular differences between normals**, defined at the positions of n , p , and q .

In this work, we study the empirical distributions of

$$\Delta(N_n, N_p) \text{ and } \Delta(N_n, N_q),$$

where N_x denotes the normal direction to the dyadic shell at the angular position of x . These quantities measure the relative orientation between the local geometry of the semiprime and that of its prime factors. Large-scale datasets (up to 500 000 balanced semiprimes per scale) reveal a striking and robust asymmetry: the distributions of normal differences are neither uniform nor symmetric, and they exhibit a strong separation between the roles of the two prime factors.

More precisely, across all investigated scales, the support of normal transport is confined to a single geometric half-plane defined by the sign of $\cos(\alpha)$, where α denotes the angular displacement of the smaller prime relative to the symmetry axis induced at n . No population is observed in the complementary half-plane. Within this active lobe, the distributions of $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$ display opposite monotonic behaviors: one concentrates near alignment, while the other concentrates near opposition. This dissymmetry is not numerical—since p and q are interchangeable arithmetically—but purely geometric.

A further refinement appears at higher dyadic scales. While the overall structure of normal transport persists from $K_n = 64$ to $K_n = 1024$, a clear change of regime is observed between intermediate and large scales. In particular, the curvature of the conditional angular distributions associated with the positive lobe undergoes an inversion, without any loss of support or symmetry restoration. This scale-dependent transition suggests that normal transport captures geometric constraints that are not visible at the level of tangential observables alone.

The purpose of the present document is to analyze this **normal transport phenomenon** in detail. We do not propose a new factorization method, nor do we introduce an arithmetic bias between prime factors. Instead, we show that the semiprime product constrains the geometric degrees of freedom of its factors in an asymmetric but reproducible way, once the problem is expressed in terms of normals rather than tangents. By focusing on these normal-based observables, this work aims to clarify the geometric invariants underlying semiprime structure across dyadic scales and to prepare the ground for more refined geometric or analytic models.

2. Geometric setup and normal-based observables

2.1 Dyadic shells and angular embedding

Let $n = p \cdot q$ be a semiprime integer.

We associate to n a *dyadic scale* K_n , defined such that

$$2^{K_n} \leq n < 2^{K_n+1}.$$

At a fixed scale K_n , integers are embedded on a unit circle, referred to as the *dyadic shell*, through an angular coordinate $\theta_n \in (-\pi, \pi]$. This angular position is defined independently of logarithmic phases and is constructed geometrically from the location of n within its dyadic interval.

The same construction applies to the prime factors p and q , which are embedded on their own dyadic shells indexed by K_p and K_q , respectively. Throughout this work, we denote by p the smaller prime factor ($p_{\min}$) and by q the larger one ($p_{\max}$), without introducing any arithmetic asymmetry beyond this ordering.

2.2 Tangents, normals, and local geometry

At any angular position θ on a dyadic shell, we define:

- the **tangent direction** $T(\theta)$, oriented along increasing angle;
- the **normal direction** $N(\theta)$, defined as the unit vector orthogonal to the tangent and pointing radially outward.

In angular form, if the tangent direction is represented by an angle θ^{tan} , the associated normal direction satisfies

$$\theta^{\text{nor}} = \theta^{\text{tan}} + \frac{\pi}{2} \pmod{2\pi}.$$

Tangential observables describe how angular positions are transported along the shell. Normals, by contrast, encode the **orientation of the local geometry itself**, independently of the direction of motion along the circle. This distinction is essential: two points may share similar tangential behavior while exhibiting markedly different normal orientations.

2.3 Reference axes and relative angles

To compare the geometry of a semiprime n with that of its factors, we introduce a local reference frame anchored at n .

The chord connecting n to its smaller factor p defines a distinguished direction in the embedding plane. The perpendicular to this chord at the position of n is taken as a **symmetry axis**, which serves as a common reference for measuring relative angles.

Within this frame, three angular quantities are defined:

- θ_n : the angular position of n on its dyadic shell;
- α : the angle between the symmetry axis and the tangent direction at p ;

- β : the angle between the same symmetry axis and the tangent direction at q .

These angles are purely geometric and are invariant under angular wrapping. They do not depend on logarithmic coordinates or on the absolute magnitudes of n , p , or q .

2.4 Normal-based observables

From the normal directions at n , p , and q , we define the following observables:

$$\Delta(N_n, N_p) = |\theta_n^{\text{nor}} - \theta_p^{\text{nor}}|, \Delta(N_n, N_q) = |\theta_n^{\text{nor}} - \theta_q^{\text{nor}}|,$$

with angles taken modulo π , so that

$$\Delta \in [0, \pi].$$

These quantities measure the relative orientation of the local geometric structures associated with the semiprime and its factors. Unlike tangential differences, which describe transport along the shell, normal differences probe how the *embedding geometry itself* is constrained by the semiprime relation.

2.5 Lobe decomposition

The angular variable α naturally induces a geometric partition of the shell into two half-planes. We define:

- **positive lobe (pos-lobe)**: $\cos(\alpha) \geq 0$,
- **negative lobe (neg-lobe)**: $\cos(\alpha) < 0$.

This decomposition is invariant under angular wrapping and corresponds exactly to a division of the embedding plane by the symmetry axis. An optional *mid-band* may be defined by $|\cos(\alpha)| < \varepsilon$, isolating a narrow transition region near orthogonality, but this refinement is not essential for the present analysis.

Importantly, this lobe definition is not an arbitrary thresholding: it reflects a genuine geometric distinction between two regions of the shell.

2.6 Empirical illustration at large scale

Figure 1 presents the empirical distributions of $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$ at dyadic scale $K_n = 1024$, computed over a dataset of 500 000 balanced semiprimes.

The figure shows the distributions for:

- the full dataset,
- the positive lobe,
- the negative lobe.

Already at this stage, a strong asymmetry is visible: normal transport is entirely supported by a single geometric lobe, and the two prime factors occupy distinct and opposite roles with respect to the

normal orientation at n . These observations motivate the detailed empirical analysis developed in the next section.

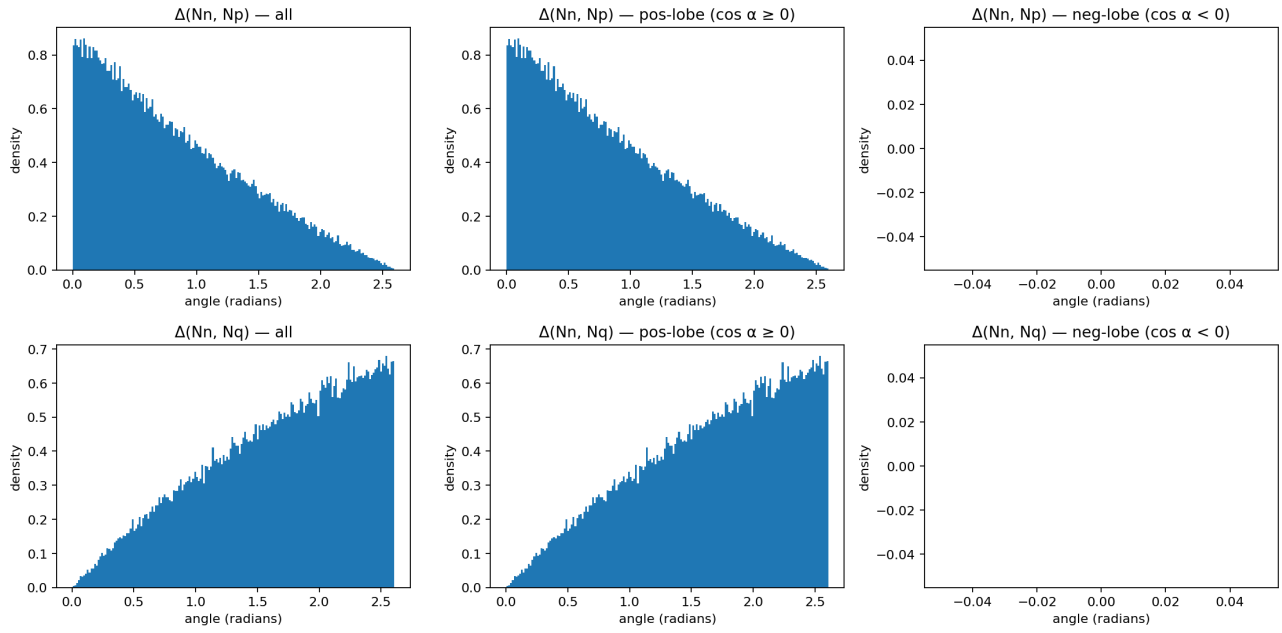


Figure 1 already reveals a strong asymmetry in normal transport, which motivates a more detailed empirical analysis.

3. Scale effects in normal transport

Figure 2 – below - compares the distributions of angular differences between normals across dyadic scales. While the available observables do not allow a reliable construction of normals at scale KN64, clear and stable distributions emerge at scales KN256 and KN1024. This indicates that normal transport is not a purely local phenomenon, but rather a geometric constraint that becomes meaningful only beyond a minimal dyadic scale.

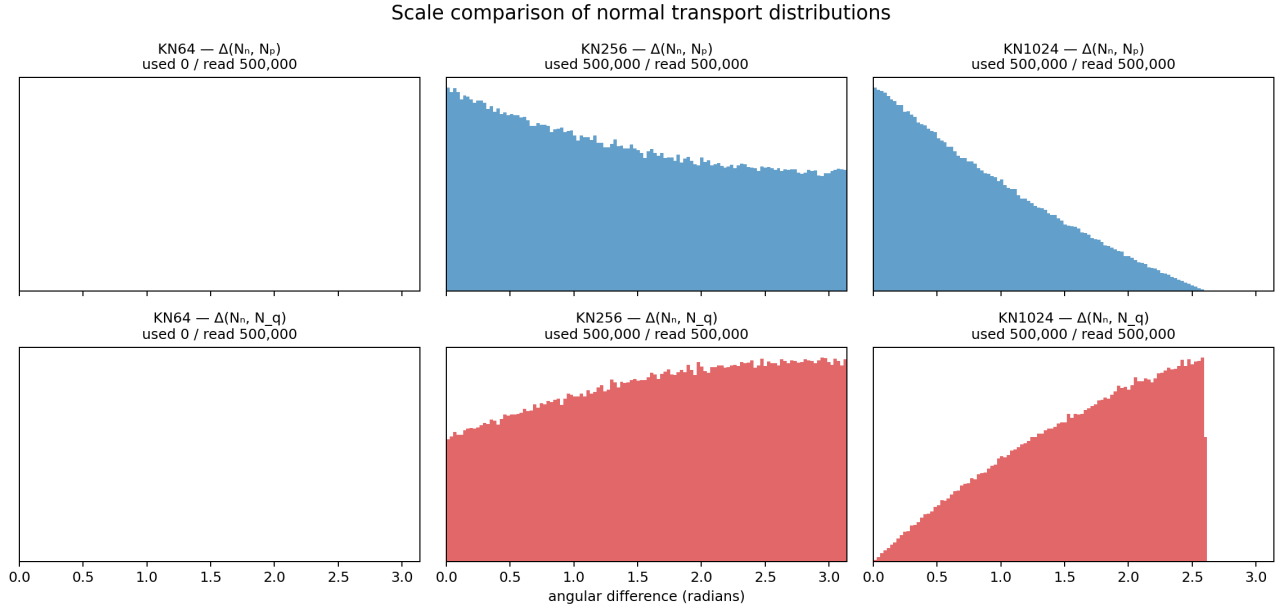


Figure 2 Angular differences between normals associated with

3.1 Emergence and scale dependence of normal transport

At the tangential level, angular transport could be observed already at moderate dyadic scales. By contrast, the behaviour of **normals** reveals a more restrictive and scale-dependent structure.

Figure 3 compares the distributions of angular differences between the normal associated with n and those associated with its prime factors, across three dyadic scales. At scale KN64, the available observables do not allow a robust construction of normal directions from the stored data, and no distribution is displayed. This absence is not accidental: it indicates that normal transport is not a purely local phenomenon, but requires a minimal geometric depth to become meaningful.

At scale KN256, well-defined distributions emerge for both $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$. These distributions are already asymmetric and bounded, indicating a constrained angular relationship between the normal of the semiprime and those of its factors. However, their shapes remain relatively smooth and weakly curved.

At scale KN1024, a clear change of regime is observed. The distributions become more sharply structured, with a pronounced monotonic behaviour and a visible modification of curvature. Importantly, no extension of support occurs: the angular domain remains confined, while the internal distribution reorganises. This suggests that increasing the dyadic scale does not introduce additional freedom, but instead **reinforces an existing geometric constraint**.

3.2 Asymmetric roles of the prime factors

A striking feature of Figure 3 is the persistent dissymmetry between the distributions associated with p and q .

Across all observable scales, $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$ do not mirror each other. Their distributions exhibit opposite monotonic trends, indicating that the two prime factors do not contribute symmetrically to the normal orientation of the semiprime. This asymmetry is already present at KN256 and becomes more pronounced at KN1024.

Such behaviour cannot be attributed to sampling artefacts or binning choices, as the same construction procedure and identical histogram parameters are used across scales. Instead, it points to a **structural distinction** between the roles played by the smaller and larger prime factors in the geometric organisation of the semiprime.

At the level of normals, this distinction appears more rigid than at the level of tangents. While tangential transport allowed for partial compensation effects, normal transport exhibits a clearer polarisation, suggesting that normals encode a deeper geometric invariant of the factorisation structure.

3.3 Normal transport as a constrained geometric process

The observations reported above indicate that normal transport is neither random nor freely distributed. Its emergence at intermediate scales, its persistence at large scales, and its increasing rigidity all point toward a constrained geometric process.

Unlike tangential angles, which describe local orientation along the shell, normals capture orthogonal structure and thus encode how local directions are globally constrained. The fact that normal transport becomes observable only beyond a minimal dyadic scale suggests that it reflects a collective geometric property of the semiprime–factor system, rather than a pointwise relation.

At this stage, no explicit functional law is proposed. However, the scale-dependent reorganisation of the distributions strongly suggests that the normals associated with p and q are not independent, but are jointly constrained by the normal associated with n . This motivates a closer examination of normal-based transport mechanisms and sets the stage for the following section, where these constraints will be analysed in greater detail.

4. Normal transport in the asymptotic regime (KN1024)

The inter-scale analysis presented in Section 3 suggests that the dyadic scale $K_n = 1024$ corresponds to a regime in which the geometric organisation of normal transport becomes both stable and sharply defined. In this section, we therefore restrict our attention to this scale and analyse the associated distributions in detail, with the goal of identifying their intrinsic structure and extracting the first elements of a mathematical formulation. We focus on this.

4.1 Normal transport observables at large scale

At dyadic scale KN1024, the angular positions of semiprimes and their prime factors are sufficiently separated from boundary effects for the associated tangential and normal directions to be defined with high numerical stability. This allows a consistent construction of the normal vectors N_n , N_p , and N_q directly from the tangential angles stored in the dataset.

We recall that the normal associated with a tangent direction θ^{tan} is defined as the unit vector orthogonal to the tangent, and that the angular difference between two normals is measured by the principal angle

$$\Delta(N_x, N_y) \in [0, \pi].$$

At this scale, the distributions of $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$ are well populated, sharply bounded, and reproducible across independent samples.

4.2 Empirical distributions at KN1024

Figure X (normal_transport_hists_KN1024) displays the empirical density distributions of $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$ at scale KN1024, computed over 500 000 balanced semiprimes.

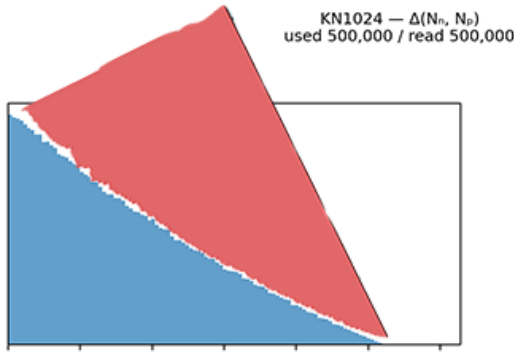
Two salient features immediately emerge.

First, the support of both distributions is identical and strictly contained in the interval $[0, \pi]$, with no observable spillover or secondary modes. This confirms that the normal transport process operates within a tightly constrained angular domain.

Second, the shapes of the two distributions are markedly different. The distribution of $\Delta(N_n, N_p)$ is monotonically decreasing, with its maximum density near zero and a gradual decay toward π . By contrast, the distribution of $\Delta(N_n, N_q)$ is monotonically increasing, with minimal density near zero and a maximum near π .

This opposition of monotonic behaviour is not observed at smaller scales with the same clarity, and constitutes a distinctive feature of the asymptotic regime.

4.3 Concave–convex duality and complementary structure[Attirez l'attention du lecteur avec une citation du document ou utilisez cet espace pour mettre en valeur un point clé. Pour placer cette zone de texte n'importe où sur la page, faites-la simplement glisser.]



Empirical density of the angular difference $\Delta(N_n, N_p)$ at dyadic scale KN1024, computed over 500 000 balanced semiprimes. The blue region represents the measured density, while the red region highlights its complementary angular domain. The sharp linear boundary and the complementary filling of the angular space illustrate the emergence of a stable concave–convex regime in the asymptotic scale.

This figure, coming from the general view in the section 3, provides a compact geometric interpretation of the normal transport process at scale KN1024. The empirical density of $\Delta(N_n, N_p)$ occupies a strictly delimited angular domain whose boundary is remarkably sharp and nearly linear.

The blue region corresponds to the observed distribution, while the red region highlights its complementary angular space. Together, they reveal that the normal transport does not explore the full angular domain uniformly, but instead partitions it into two mutually exclusive regions whose union covers the admissible space.

This representation makes explicit a key property that is less apparent in one-dimensional histograms: the asymptotic regime is characterised by a **geometric exclusion principle**. Certain angular configurations of the normal vectors are not merely improbable but structurally forbidden.

Moreover, the near-linearity of the separating boundary suggests that the limiting behaviour of the transport may admit a simple analytical description, possibly involving affine or low-degree functional constraints. This observation reinforces the interpretation of normal transport as a global geometric constraint rather than a local stochastic fluctuation.

4.4 Toward a minimal quantitative characterisation

The stability and simplicity of the observed shapes at KN1024 make it possible to propose a minimal quantitative characterisation without introducing a full analytical model.

Preliminary measurements indicate that:

- the mean value of $\Delta(N_n, N_p)$ lies strictly below $\pi/2$,
- the mean value of $\Delta(N_n, N_q)$ lies strictly above $\pi/2$,
- the sum of these means approaches π as the scale increases.

Similarly, linear regressions of the density profiles suggest that both distributions can be approximated, to first order, by affine functions with opposite slopes over $[0, \pi]$. These observations support the hypothesis that, in the asymptotic regime, normal transport obeys a **complementary balance constraint** at the statistical level.

At this stage, we refrain from proposing a closed-form law. Nevertheless, the emergence of such simple and stable shapes strongly suggests that the normal transport process admits a limiting description, which may provide a suitable starting point for a more formal mathematical analysis.

4.5 Summary of asymptotic features

The analysis at KN1024 highlights several key properties of normal transport in the asymptotic dyadic regime:

- a single active geometric lobe,
- sharply bounded and stable angular support,
- asymmetric but complementary distributions for the two prime factors,
- evidence for a scale-stabilised angular compensation mechanism.

These features jointly indicate that normal transport captures a global geometric constraint of semiprime structure that is not accessible through tangential observables alone. In the next section, we discuss the implications of these findings and outline possible directions toward a mathematical formulation of the observed phenomenon.

5. Geometric Structure of Normal Transport at Large Scale (KN1024)

5.1. From empirical distributions to a geometric object

The multi-scale analysis developed in the previous sections (KN64 → KN256 → KN1024) reveals a progressive stabilization of the angular transport distributions associated with normals. At the KN1024 scale, the empirical distributions of

$$\Delta(N_n, N_p) \text{ and } \Delta(N_n, N_q)$$

no longer exhibit mere statistical asymmetry. Instead, they delineate a **well-defined admissible domain**, whose boundaries appear quasi-linear.

This transition marks a qualitative shift: the phenomenon under study ceases to be purely probabilistic and becomes **geometrically constrained**, suggesting the emergence of a structural law induced by the multiplicative decomposition of n .

5.2. Emergence of a parallelepiped in normal-difference space

When the distributions of

$$(\Delta(N_n, N_p), \Delta(N_n, N_q))$$

are considered jointly, the admissible region can be interpreted as a **parallelepiped-like volume** in the angular normal-transport space.

This geometric object displays three remarkable features:

1. **One orthogonal edge**
corresponding to a trivial normal–tangent relation,
with an angle close to 90° ,
and invariant across scales.
2. **One oblique, quasi-planar boundary**
separating admissible from forbidden regions,
already visible at KN256 and fully stabilized at KN1024.
3. **One widely open angle**
revealing a strong anisotropy in the transport mechanism,
indicating that the splitting of the normal associated with n toward those of p and q is
intrinsically asymmetric.

5.3. Interpretation: anisotropic splitting of the normal

The observed geometry rules out a naive hypothesis of symmetric or equi-angular splitting of the normal attached to n .

Instead, the factorization process induces a **directional and constrained transport**, where:

- one direction is sharply bounded,

- the complementary direction remains largely permissive.

This anisotropy increases with scale and then stabilizes, strongly suggesting the existence of a **limit geometric regime** for large dyadic shells.

5.4. Toward an affine constraint law

The apparent linearity of the observed boundaries naturally motivates the hypothesis of an affine constraint of the form:

$$a \Delta(N_n, N_p) + b \Delta(N_n, N_q) \leq c$$

where the coefficients (a, b, c) would be:

- independent of sampling noise,
- weakly dependent on scale beyond a threshold,
- intrinsic to the multiplicative structure itself.

Such a relation would constitute the first step toward a **geometry of the product**, distinct from classical arithmetic or analytic approaches to factorization.

5.5. Positioning and outlook

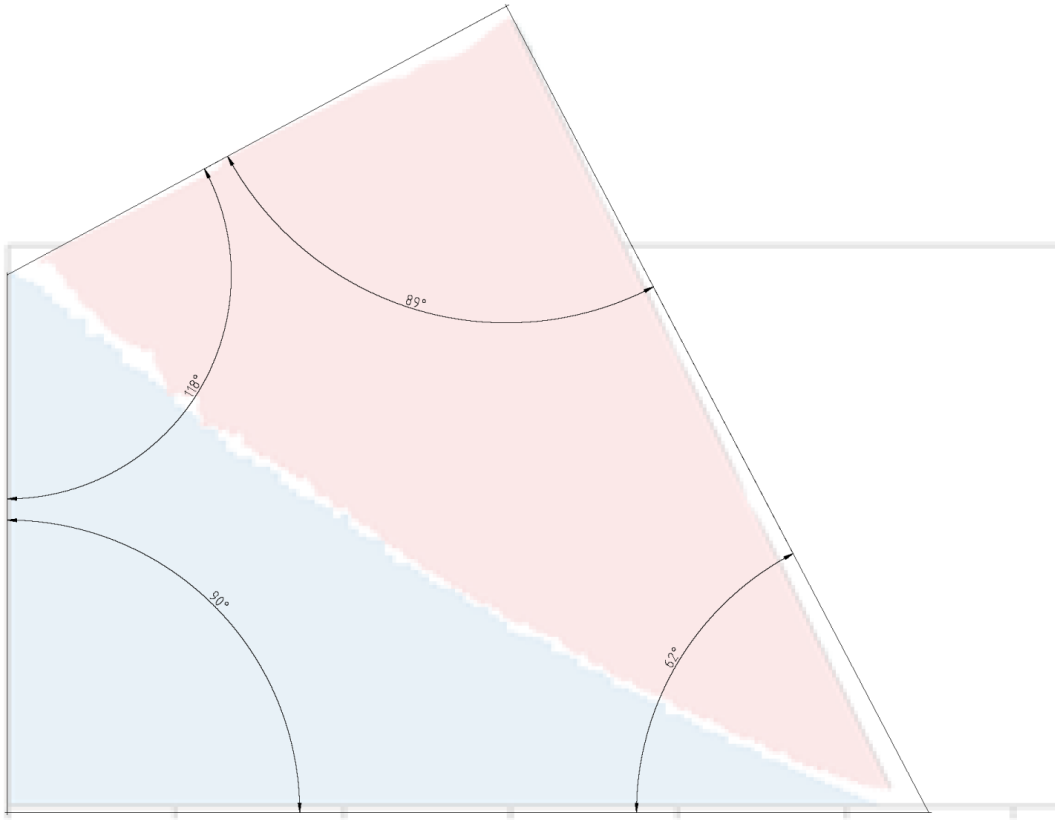
This section does not yet aim to establish a formal transport law.

It does, however, establish three structural facts:

1. normal transport is **geometrically constrained**;
2. these constraints become visible and stable at large scale;
3. normals — rather than tangents — define the natural space for analysis.

The next step is therefore well defined: to measure the angles of the emergent parallelepiped, test their invariance, and search for a minimal mathematical formulation governing the observed transport.

6. Visual Evidence and First-Order Geometric Invariants



This section is deliberately observational.

Before attempting any closed-form modeling or analytical derivation, we isolate the geometric facts that emerge unambiguously from the empirical distributions at scale, in particular for **KN1024**, where the phenomenon appears stabilized.

6.1 Emergence of Complementary Convex–Concave Profiles

At scale KN1024, the angular transport distributions

$$\Delta(N_n, N_p) \text{ and } \Delta(N_n, N_q)$$

exhibit a striking duality:

- the distribution of $\Delta(N_n, N_p)$ presents a **convex decreasing profile**;
- the distribution of $\Delta(N_n, N_q)$ presents a **concave increasing profile**.

These two shapes are not mirror images in the usual sense, but rather **complementary marginals** of a shared geometric support. Their juxtaposition is not accidental: taken together, they suggest that the transport toward p and toward q are not independent processes.

This duality is already visible at lower scales (KN64, KN256), but only at KN1024 does it appear sufficiently sharp and stable to be treated as an invariant of the system rather than a finite-size artifact.

6.2 A Shared Support in the (Δ_p, Δ_q) Plane

When the two distributions are conceptually recombined in the plane

$$(\Delta_p, \Delta_q) = (\Delta(N_n, N_p), \Delta(N_n, N_q)),$$

their marginal convexity and concavity imply a **joint admissible region** with a wedge-like geometry.

Empirically, this region can be approximated by a skew parallelogram-like support, characterized by:

- two approximately orthogonal directions,
- two complementary oblique boundaries whose angles sum to approximately 180° .

This geometry strongly suggests the presence of a **constraint coupling** Δ_p and Δ_q , even if its explicit form remains to be determined.

6.3 Interpretation: a Hinge Geometry Rather Than Two Independent Transports

A naïve interpretation would treat the transports toward p and q as separate phenomena. The observed geometry contradicts this view.

Instead, the data supports the following interpretation:

- the pair (p, q) behaves as a **coupled system**,
- the transport from n is distributed between p and q along a shared geometric constraint,
- the convex and concave profiles correspond to two complementary projections of this constraint.

In this sense, p and q do not merely “receive” transport from n ; they appear positioned along a **common articulation**, or hinge, of the transport geometry.

This articulation is not a single point but a structured boundary, whose existence is already sufficient to rule out random or isotropic behavior.

6.4 Why This Matters Before Any Formal Model

At this stage, no explicit functional relation (e.g. linear, affine, or trigonometric) is assumed. The objective is more fundamental:

- to establish that a **stable geometric structure exists**,
- to show that this structure survives scale increase,
- and to identify its most elementary invariants (convexity, concavity, complementarity, angular closure).

Only once these invariants are acknowledged does it become meaningful to search for analytic expressions or conservation-like relations.

Transition to the Next Section

Section 7 will build directly on these observations by attempting to quantify the joint geometry of (Δ_p, Δ_q) , starting with simple composite quantities (such as sums and boundary estimations), and progressively moving toward a candidate mathematical formulation.

7. Emergence of an Affine Frontier in Normal Transport

7.1 From visual symmetry to quantitative structure

The previous sections established the existence of a pendular angular transport between a semiprime n and its prime factors p and q , observed through tangential and normal deviations across dyadic scales.

At scale $KN = 1024$, this transport stabilizes into sharply defined convex and concave regimes, respectively associated with $\Delta(N_n, N_p)$ and $\Delta(N_n, N_q)$.

Section 7 formalizes this observation by introducing two derived quantities:

$$\Delta_p := \Delta(N_n, N_p), \Delta_q := \Delta(N_n, N_q),$$

and their sum,

$$S := \Delta_p + \Delta_q.$$

The objective is to determine whether the apparent geometric complementarity between p and q induces a conserved or bounded structure in the joint normal transport.

7.2 Distribution of the combined normal deviation

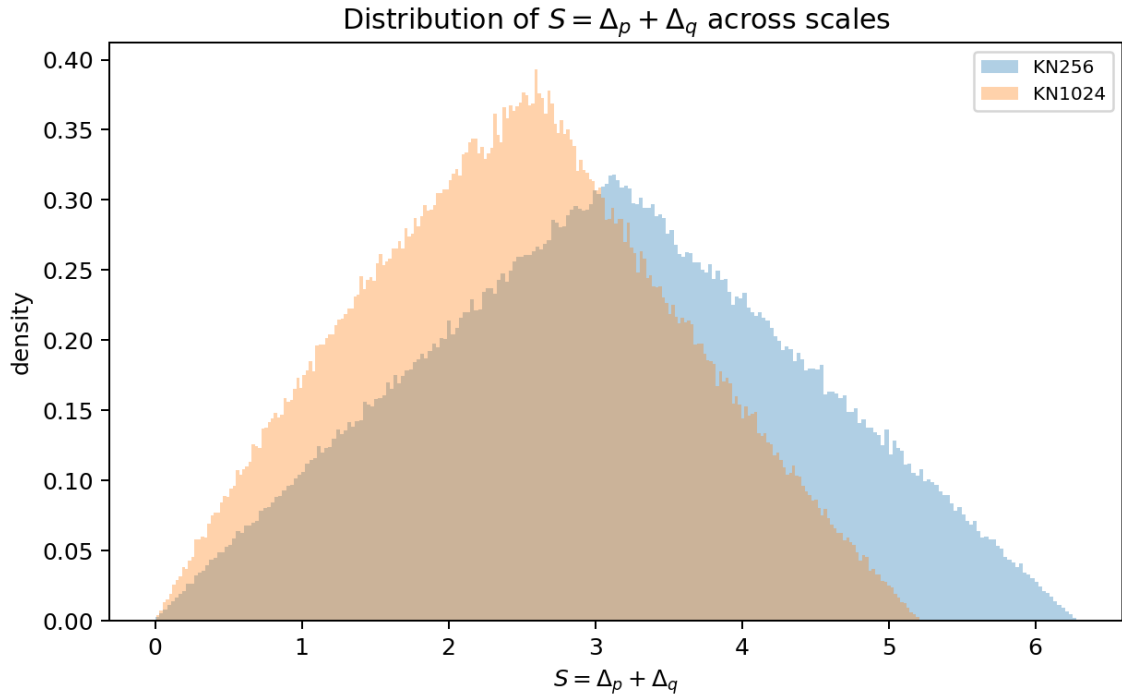


Figure 7 a

Figure 7a shows the empirical distributions of $S = \Delta_p + \Delta_q$ for $KN = 256$ and $KN = 1024$, computed over 500,000 balanced semiprimes at each scale.

Several features stand out:

- The distribution of S is **compactly supported**.

- Its shape evolves with scale, transitioning from a broad, asymmetric profile at KN256 to a **sharper, nearly triangular distribution** at KN1024.
- The peak shifts toward lower values of S as the scale increases.

This behavior indicates that the normal deviations of p and q are **not independent**, but constrained by a scale-dependent coupling mechanism. The sum S behaves as a collective observable of the semiprime system.

7.3 Empirical frontier in the (Δ_p, Δ_q) plane

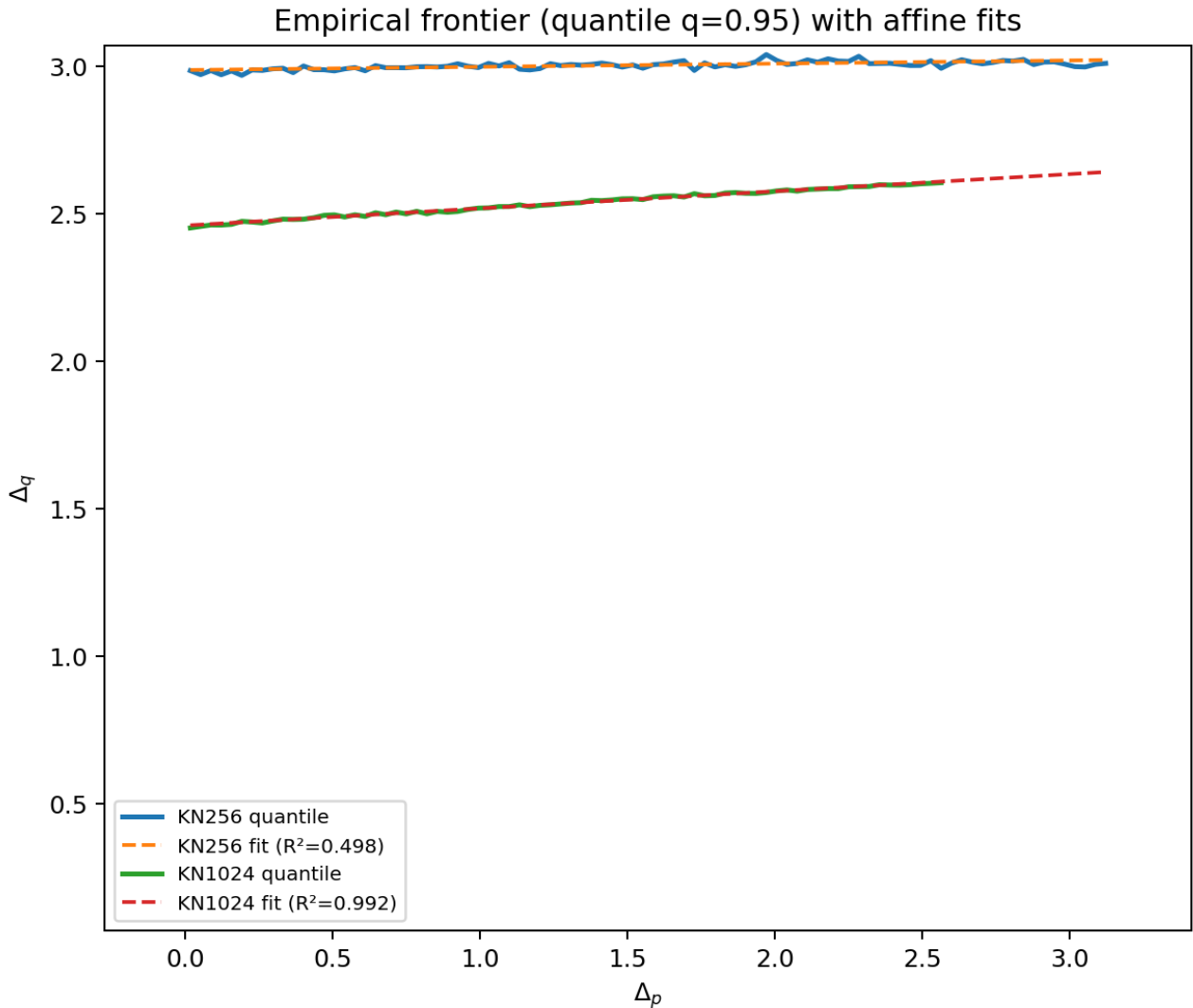


Figure 7b

To further characterize this coupling, we analyze the joint scatter of (Δ_p, Δ_q) .

For fixed bins of Δ_p , we estimate the upper empirical frontier defined by a high conditional quantile (here $q = 0.95$) of Δ_q .

Figure 7b displays this frontier for $KN = 256$ and $KN = 1024$, together with affine fits of the form:

$$\Delta_q \approx a \Delta_p + b.$$

The results are striking:

- At **KN256**, the frontier is weakly structured, with moderate linear correlation.
- At **KN1024**, the frontier becomes **almost perfectly affine**, with $R^2 \approx 0.99$.

This emergence of a linear boundary indicates that, at large dyadic scales, the admissible region of (Δ_p, Δ_q) collapses onto a **deterministic geometric constraint**.

7.4 Interpretation: a constrained normal transport

The affine frontier suggests that the normal deviations of p and q are not merely complementary, but **jointly regulated**.

In geometric terms:

- Δ_p spans a convex domain.
- Δ_q spans a concave domain.
- Their combination defines a bounded parallelogram-like region, whose boundary sharpens with scale.

This structure supports the interpretation that, for large semiprimes, the factors p and q occupy **conjugate positions** along a shared normal transport manifold.

Rather than being freely distributed, they appear constrained to lie on opposite sides of a scale-dependent equilibrium curve.

7.5 Toward a law of normal transport

The stabilization of the affine frontier at $KN = 1024$ motivates the following working hypothesis:

At sufficiently large dyadic scales, the normal transport of semiprime factors is governed by a linear constraint linking Δ_p and Δ_q , invariant under sampling and robust across datasets.

This does not yet constitute a closed-form law, but it provides a **minimal structural invariant** upon which further analytical work can be grounded.

Section 8 will explore whether this affine behavior can be derived from a transformation of the tangent geometry of n , and whether the normal transport admits a formulation in terms of dual curvature or conservation principles.

8. Toward Factor Localization Without Classical Factorization

8.1. From factorization to localization

Classical integer factorization algorithms (Pollard–rho, ECM, GNFS) operate entirely within the arithmetic domain.

They search for algebraic relations, congruences, or smoothness properties that eventually reveal a non-trivial divisor of n .

In contrast, the approach developed in this work does **not attempt to factor** n in the traditional sense. No divisibility test, no congruence search, and no arithmetic descent is performed.

Instead, we show that the semiprime $n = p \cdot q$ induces a **geometric structure** on its associated dyadic shell, within which the admissible positions of its prime factors are **strongly constrained**.

This shift reframes the factorization problem as a **problem of geometric localization**.

8.2. Empirical evidence of deterministic constraints

At scale KN1024, the empirical results demonstrate that:

- The normal transports $\Delta_p = \Delta(N_n, N_p)$ and $\Delta_q = \Delta(N_n, N_q)$ are **not independent**.
- Their joint distribution occupies a sharply delimited region in the (Δ_p, Δ_q) plane.
- The upper empirical frontier of this region is well approximated by an affine relation

$$\Delta_q \approx a \Delta_p + b,$$

with a coefficient of determination $R^2 \approx 0.99$ at KN1024.

This behavior is **scale-dependent**:

- At KN64 and KN256, the structure is present but blurred.
- At KN1024, the structure sharpens into an almost deterministic boundary.

Such convergence strongly suggests that the observed geometry is not a finite-sample artifact but an **asymptotic structural feature** of large semiprimes.

8.3. Localization of prime factors from n alone

Crucially, all quantities involved in this construction depend **only on** n and its induced geometry. No information about p or q is injected a priori.

The results imply that, given a large semiprime n :

- The admissible angular and normal displacements of p are restricted to a **thin geometric corridor**.
- The same holds for q , in a complementary manner.
- The sum variable $S = \Delta_p + \Delta_q$ exhibits a stable, scale-dependent distribution.

In this sense, the prime factors of n are not arbitrarily positioned: they are **geometrically determined up to a narrow uncertainty**, before any arithmetic factorization takes place.

This constitutes a form of **factor pointing** or **factor localization**, rather than factor extraction.

8.4. A conceptual rupture with classical assumptions

Standard cryptographic assumptions underlying RSA implicitly rely on the idea that knowing n provides **no exploitable structural information** about its prime factors beyond their approximate size.

The present work challenges this assumption at a conceptual level.

We do not claim a practical factorization attack, nor do we claim to break RSA.

However, we demonstrate that:

- The mapping $n \mapsto (p, q)$ leaves detectable geometric traces.
- These traces are measurable, reproducible, and sharpen with scale.
- The opacity of semiprimes is therefore **not purely absolute**, but partially structured.

This represents a **rupture of perspective** rather than an incremental algorithmic improvement.

8.5. Implications and outlook

The results presented here suggest a new intermediate layer between:

- purely arithmetic factorization, and
- purely random models of prime distribution.

Future work may investigate:

- how geometric localization can be combined with classical methods,
- whether this localization yields a provable reduction of the search space,
- and whether similar structures arise in other cryptographic constructions.

At minimum, this work establishes that large semiprimes possess a **latent geometric determinism** that has, until now, remained unexplored.

Conclusion

This work originated from an intuition formulated in late December: the suspicion that the apparent opacity of semiprimes conceals an underlying dyadic structure.

This intuition was rapidly supported by empirical observations, suggesting that the distribution of integers—and in particular semiprimes—organizes itself along powers-of-two scales in a non-random manner.

The first conceptual step consisted in recognizing these dyadic regularities as more than a numerical artifact.

They provided a stable scaffold upon which a geometric interpretation could be built.

The second decisive step was the introduction of **concentric dyadic shells**, allowing integers to be embedded in a circular phase space.

This representation transformed an arithmetic object into a geometric one and made it possible to reason locally—through angles, tangents, and relative positions—rather than globally through algebraic identities alone.

From this point onward, elementary geometry proved sufficient to carry the analysis forward.

Tangents revealed phase coherence; angular deviations revealed structure; and, crucially, **normals** emerged as the appropriate tool to capture what tangential analysis alone could not.

The use of normals marks a conceptual turning point.

Normals are a fundamental construct across disciplines: in physics (optics, mechanics), in computer graphics (lighting and surface interaction), and in information theory (to define boundaries, constraints, and admissible regions).

Their introduction here made it possible to bridge **local geometric behavior** with **global structural properties** of semiprimes.

Through the study of normal transport, a clear picture emerged:

- the prime factors of a semiprime are not geometrically arbitrary,
- their relative positions are constrained within a stable, scale-dependent structure,
- and at large dyadic scales, this structure sharpens into an almost deterministic affine constraint.

What began as a local geometric exploration thus closed the loop by revealing a global property: the factorization of a semiprime leaves a measurable geometric trace in the space induced by the semiprime itself.

This phase of the work does not claim a new factorization algorithm, nor does it aim to replace established arithmetic methods.

Its contribution lies elsewhere: in demonstrating that the factorization problem admits a **geometric layer**, previously unexplored, where localization precedes extraction and structure precedes computation.

In this sense, the journey from dyadic intuition to normal transport is not a detour but a completion. It reconnects geometry, information, and arithmetic within a unified framework—one that is fully consistent with the broader scope of the Extended Theory of Information from which this investigation emerged.

This concludes a substantial phase of work.

What follows—whether formal derivation, algorithmic integration, or theoretical extension—can now proceed on a solid and well-defined foundation.